# News Perplexity*

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**Abstract**

Existing research suggests that investors inefficiently process text that induces cognitive strain. This paper challenges that view. Drawing on dual-process theory, I hypothesize that certain strain-inducing text disrupts heuristic (System 1) intuition and triggers deliberate (System 2) analysis. To capture such text, I introduce news perplexity—a metric derived from large language models—and find that high perplexity news significantly mitigates market underreaction. Validating the mechanism, high perplexity news coincides with a spike in EDGAR search traffic, which is consistent with the notion that investors respond to the cognitive trigger by acquiring more information. These findings provide the first evidence that beneficial cognitive strain enhances market efficiency by forcing deeper information processing.

**JEL Classification:** C55, D83, G12, G14, G41

**Keywords**: Cognitive strain, Investor information processing, Large language models, Market underreaction

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# 1 Introduction

How investors process information is central to asset pricing. The prevailing view in empirical finance holds that cognitive strain impedes information processing—a friction that exacerbates limited attention and delays price discovery. The literature extensively documents that complexity and low readability act as barriers, causing investors to avoid dense reports (Li, 2008; Loughran and McDonald, 2014; Bonsall, Leone, Miller, and Rennekamp, 2017; Miller, 2010) and underreact to information that is difficult to process (Hirshleifer and Teoh, 2003; Hirshleifer et al., 2009; You and Zhang, 2009).

This paper challenges that consensus, not by disputing these findings, but by arguing that they capture only one dimension of cognitive strain. Drawing on dual-process theory (Kahneman, 2011; Evans and Stanovich, 2013), I posit that cognitive strain is not merely a barrier, but can function as a beneficial signal. Dual-process theories distinguish between Type 1 processing—fast, automatic, and intuitive—and Type 2 processing—slow, deliberate, and analytical (Stanovich and West, 2000; Frederick, 2005). A critical question is what causes the shift from the default Type 1 to the more costly Type 2 mode. Type 1 generates default responses that Type 2 can override, but overriding requires both the cognitive capacity for deliberate reasoning and awareness that such reasoning is needed. When investors encounter predictable information flows, they rely on fast, heuristic processing; when they encounter unexpected narratives, the breakdown of processing fluency signals that default responses may be inadequate, triggering analytical engagement.

This mechanism has profound implications for understanding market efficiency. Financial news media serves as a primary channel through which information reaches investors (Tetlock, 2007), yet the persistence of delayed price adjustments following news—the well-documented underreaction anomaly (Ball and Brown, 1968; Bernard and Thomas, 1989, 1990; Chan et al., 1996)—suggests that prices often fail to immediately

incorporate available information. While standard explanations attribute this inefficiency to limited attention (Dellavigna and Pollet, 2009; Hirshleifer et al., 2009; Barber and Odean, 2008) or cognitive constraints (Peng and Xiong, 2006; Hong and Stein, 1999), I propose that the strain-inducing properties of the text are an important determinant of how cognitive resources are allocated. If strain-inducing text triggers deeper processing, it should accelerate price discovery and mitigate underreaction.

To test this hypothesis, I introduce *news perplexity*, a metric derived from Large Language Models (LLMs) that captures the narrative unpredictability of financial news. Perplexity measures how "surprised" a language model is by a sequence of text—formally, the exponentiated average negative log-likelihood of the realized words given preceding context (Shannon, 1948; Bengio et al., 2003). This measure is theoretically distinct from traditional readability metrics like the Fog Index, which capture syntactic complexity through surface-level features such as sentence length and syllable counts (Gunning, 1968; Li, 2008). A text can be syntactically simple (low Fog) yet narratively unpredictable (high perplexity), or syntactically complex (high Fog) yet stylistically conventional (low perplexity). This distinction is crucial: while syntactic complexity impedes processing through decoding costs, narrative unpredictability induces cognitive strain by violating probabilistic expectations—and it is this violation that triggers the shift to System 2 processing.

The theoretical connection between perplexity and cognitive strain rests on the "predictive brain" hypothesis from cognitive neuroscience (Clark, 2013; Friston, 2010). This framework conceptualizes the brain as a prediction machine that continuously generates expectations about incoming stimuli and learns from prediction errors. High perplexity text—text that deviates from expected patterns—generates prediction errors that signal the inadequacy of default processing strategies. Just as an LLM assigns low probability to unexpected word sequences, an investor's cognitive system experiences "surprise" when narratives violate linguistic expectations, prompting a reallocation of attention and deeper

analysis (Rao and Ballard, 1999). Perplexity thus provides a computable proxy for the cognitive jolt that disrupts heuristic processing.

A critical methodological challenge in applying LLMs to historical finance studies is lookahead bias. Standard models such as GPT-5 are trained on internet corpora that include text written after the period being studied—not only future outcomes of historical events, but also evolved linguistic conventions, terminology, and writing styles. This risks contaminating perplexity measures with linguistic patterns unavailable to investors at the time. To overcome this, I employ the chronologically consistent models of He, Lv, Manela, and Wu (2025)—ChronoGPT. These time-stamped models are trained only on text available through each historical period, ensuring that my perplexity measure reflects the genuine linguistic surprise of narratives as investors would have experienced them ex ante.

Using a sample of high-impact corporate news from the Dow Jones Newswire (2008–2023), I find strong evidence that strain-inducing text benefits market efficiency. My central finding is that high perplexity news significantly mitigates market underreaction. Specifically, while stock prices drift gradually for weeks following low-perplexity news, they adjust almost immediately following high-perplexity news. I quantify this using the Delayed Response Ratio (Dellavigna and Pollet, 2009)—the ratio of the delayed price divergence between positive and negative news to the total price divergence. This ratio stands at 40% for low perplexity news but falls to just 11% for high perplexity news. The 29 percentage point reduction constitutes a statistically and economically significant attenuation of underreaction. This result is robust to extensive controls: day-of-week effects, stock characteristics (volatility, illiquidity, market capitalization, news coverage volume), industry fixed effects, content controls (predicted long-term returns from LLM embeddings, following Chen, Kelly, and Xiu, 2022), syntactic measures (article length, Fog Index), and news subject fixed effects.

To validate the proposed mechanism—that high perplexity triggers actual analytical effort rather than mere confusion—I examine abnormal traffic to the SEC's EDGAR database, a validated proxy for costly, deliberate information acquisition (Drake et al., 2015; Loughran and McDonald, 2017). If strain-inducing text engages System 2 processing, investors should actively seek out primary sources to verify and analyze information. I find that high perplexity news is associated with a sharp spike in EDGAR search volume—an incremental increase of 35% of the magnitude of the baseline low-perplexity group—even after controlling for general attention proxied by Google Search Volume Index (Da et al., 2011; deHaan et al., 2023). This finding suggests that strain-inducing text drives deliberate information acquisition, providing a micro-founded explanation for the accelerated price discovery.

This paper contributes to three strands of the literature. First, my contribution to behavioral finance is to provide the first evidence that cognitive strain induced by financial news text can be beneficial for market efficiency. The existing literature predominantly finds that complexity and low readability act as barriers, causing investors to avoid dense reports (Li, 2008; Loughran and McDonald, 2014; Bonsall, Leone, Miller, and Rennekamp, 2017; Miller, 2010) and underreact to information that is difficult to process (Hirshleifer and Teoh, 2003; Hirshleifer et al., 2009; You and Zhang, 2009). I introduce a distinct dimension of strain-inducing text: narrative unpredictability, defined as the surprise elicited when text violates probabilistic expectations, which I measure using LLM perplexity. Drawing on dual-process theory, which posits that cognitive strain can disrupt heuristic processing and engage analytical thinking (Kahneman, 2011; Evans and Stanovich, 2013), I provide empirical evidence that this specific form of text-induced strain acts not as a barrier, but as a beneficial trigger that prompts deeper information processing. This provides a novel cognitive channel for understanding the persistence of market underreaction.

Second, I contribute to the measurement of textual properties in asset pricing by

conceptualizing LLM perplexity as a proxy for the cognitive strain induced by narrative unpredictability—the mental jolt experienced when text deviates from expected linguistic patterns, disrupting automatic processing (Ilut and Valchev, 2023). While prior work has linked unusual news content to market outcomes—notably, Glasserman and Mamaysky (2019) use corpus-based n-gram frequencies to predict future volatility—my contribution is distinct. Perplexity, unlike corpus-based methods, captures richer contextual unpredictability by leveraging the sophisticated probabilistic models of language learned by LLMs. Crucially, I distinguish narrative unpredictability both from the underlying economic content of the news (e.g., textual sentiment: Tetlock, 2007) and from syntactic complexity (readability: Li, 2008). By controlling for predicted economic impact and readability, I isolate the effect of the text's narrative structure, demonstrating that how information is conveyed—not just what is conveyed—is a distinct determinant of how investors process information.

Third, I contribute to the methodological application of LLMs in finance. This paper is the first to employ LLM perplexity to explain the persistence of market underreaction. The rapidly growing literature applying LLMs in finance typically focuses on prediction (Lopez-Lira and Tang, 2023; Chen, Kelly, and Xiu, 2022) or refining existing textual measures (Zang, Zheng, and Zheng, 2024). My approach demonstrates a novel, theory-driven use of an intrinsic LLM property—its probabilistic measure of surprise—to quantify a cognitive construct. By employing chronologically consistent models (He, Lv, Manela, and Wu, 2025), I ensure that perplexity scores are free from lookahead bias, enabling credible causal inference about investor behavior. This methodological innovation highlights the potential of LLMs not just for prediction, but for measurement and explanation in asset pricing research.

The remainder of the paper is organized as follows. Section 2 describes the construction of the perplexity measure and the data. Section 3 presents the empirical results, examining the effect of perplexity on market underreaction and providing evidence for the cognitive

mechanism using EDGAR search data. Section 4 concludes.

# 2 Methodology and Data

## 2.1 Perplexity as a Measure of Cognitive Strain

### 2.1.1 Why Not Readability?

To empirically test the hypothesis that strain-inducing text engages more deliberate System 2 processing, I must first operationalize the cognitive demand of the text. Prior literature has relied on readability metrics such as the Gunning Fog Index (Gunning, 1968), popularized in finance by Li (2008) and extended by measures like the Bog Index (Bonsall, Leone, Miller, and Rennekamp, 2017). While useful, these metrics are ill-suited for my purposes. They primarily capture the cost of *decoding*—sentence length, word complexity, passive voice—rather than the *predictability* that induces cognitive strain in information processing. A text can be syntactically simple yet stylistically unpredictable, or syntactically complex yet conveying information in a formulaic, boilerplate fashion.

Loughran and McDonald (2014) document serious problems with the Fog Index in financial applications: complex words (three or more syllables) in business documents are often industry-specific terms rather than genuinely challenging vocabulary, and the two components of Fog (complex words and sentence length) are negatively correlated in their sample. More fundamentally, readability metrics focus on surface features that affect comprehension *cost*—syntactic complexity may impede processing through decoding costs. My hypothesis, by contrast, concerns narrative unpredictability: cognitive strain arising from violated probabilistic expectations, the *surprise* that triggers deeper processing.

### 2.1.2 Perplexity and the Predictive Brain

My hypothesis requires a measure that quantifies the degree of cognitive effort or "narrative surprise"—the extent to which a narrative violates probabilistic expectations. For this, I employ perplexity, a standard measure in information theory for evaluating how well a probability model predicts a sample (Shannon, 1948; Cover, 1999). An LLM, trained on a vast corpus, develops a sophisticated probabilistic model of language, learning its typical patterns, associations, and structures (Bengio et al., 2003; Vaswani et al., 2017). When computed using an LLM, perplexity measures how "surprised" the model is by a new sequence of words—effectively quantifying the predictive difficulty of the text.

Formally, for a text sequence $W$ consisting of $N$ tokens $(w_1, w_2, ..., w_N)$, perplexity is defined as the exponentiated average negative log-likelihood of the sequence:

$$\text{Perplexity}(W) = exp\left(-\frac{1}{N}\sum_{i=1}^{N}\log P(w_i|w_1,\ldots,w_{i-1})\right) \tag{1}$$

where $P(w_i|w_1,...,w_{i-1})$ is the conditional probability assigned by the model to the token $w_i$ given the preceding tokens. Text that is predictable (high probability) will have low perplexity, while text that violates expectations (low probability) will have high perplexity. Perplexity is mathematically equivalent to the exponentiation of cross-entropy between the model's predicted distribution and the actual text sequence. It can be interpreted as a "branching factor"—roughly the number of equally probable words that could follow any given context. A perplexity of 50 means the model is, on average, as uncertain as if choosing among 50 equally likely options at each position.

The theoretical link between LLM perplexity and human cognitive processing rests on the "predictive brain" hypothesis (Clark, 2013; Friston, 2010). This framework conceptualizes the brain as a prediction machine that continuously generates expectations about incoming stimuli. Perception, in this view, is not passive reception but active inference:

the brain maintains a generative model of the world and compares predictions against sensory input (Rao and Ballard, 1999). Prediction errors—mismatches between expected and actual stimuli—drive learning, attention allocation, and behavioral adjustment. Just as an LLM assigns probabilities to word sequences based on learned patterns, the human brain generates predictions about linguistic input based on experience. When text conforms to expectations (low perplexity), processing proceeds fluently via automatic, Type 1 mechanisms. When text violates expectations (high perplexity), prediction errors accumulate, signaling that default responses may be inadequate and triggering the engagement of controlled, Type 2 processing.

Critically, in my empirical design, I control for the underlying economic content and topic of the news. Therefore, perplexity does not measure informational surprise (the magnitude of the economic event itself), but rather *narrative* unpredictability—the deviation of phrasing, structure, and narrative from expected patterns. A high perplexity score indicates that the *way* information is conveyed deviates from linguistic norms, independent of *what* information is conveyed. This makes perplexity an ideal proxy for the cognitive jolt that disrupts automatic processing and triggers analytical engagement.

#### 2.1.3 Illustrative Examples

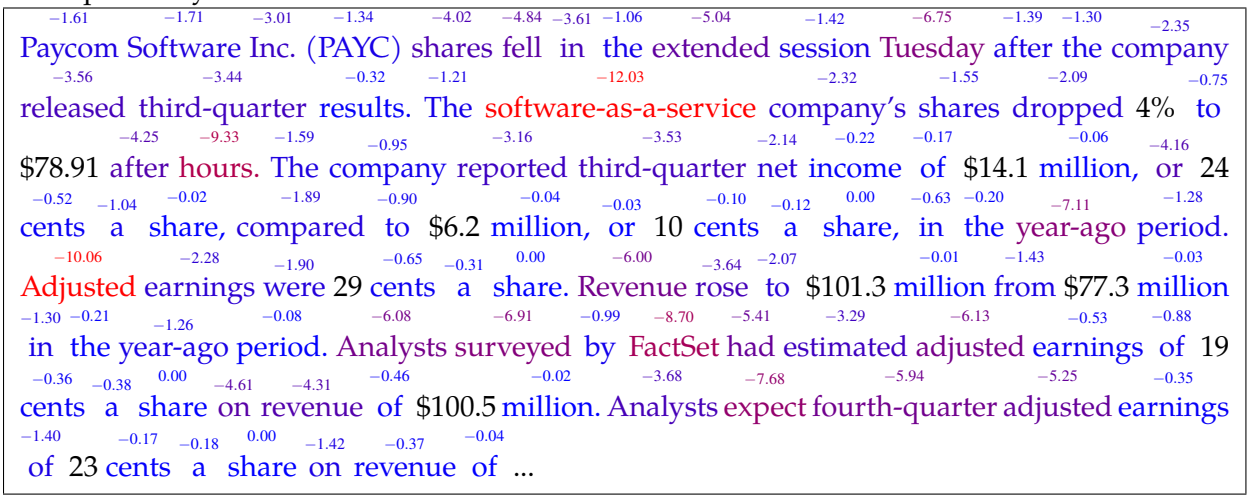
To illustrate the distinction between low and high perplexity text, I present four examples from my sample. Each word is annotated with its log probability from the language model, with lower (more negative) values indicating greater surprise. Words highlighted in red have log probabilities below $-10$, indicating extreme unpredictability. The color gradient from blue to purple to red reflects increasing surprise.

The first two examples represent **low perplexity** articles—text that follows predictable, formulaic patterns typical of standard financial news reporting. The second two examples represent **high perplexity** articles—text that employs unexpected metaphors, colorful

language, and narrative structures that deviate from conventional financial journalism.

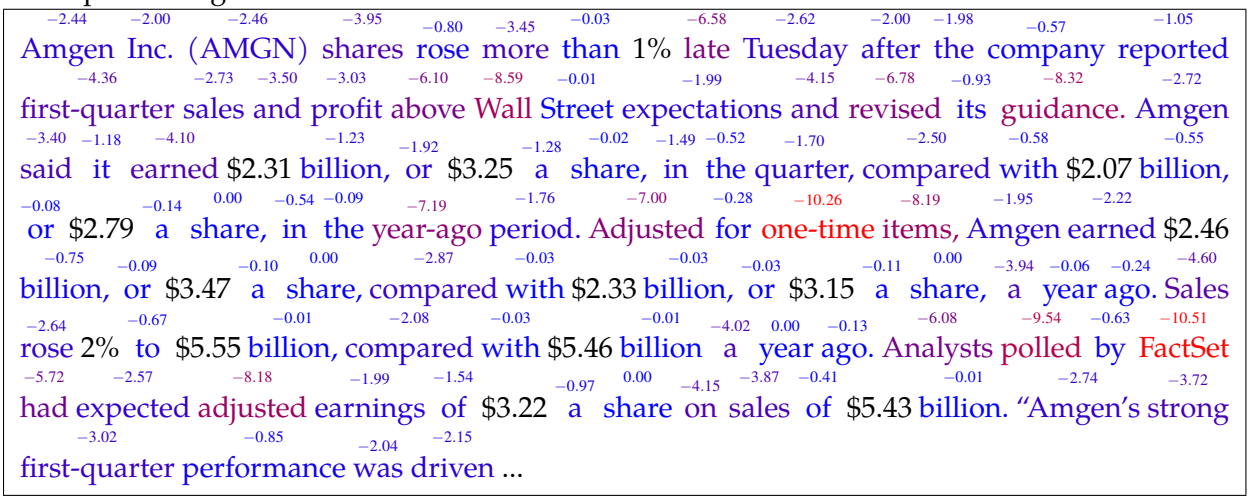
**Low Perplexity Examples**

Example 1: Paycom Software




This Paycom Software article exemplifies formulaic earnings reporting. The structure follows a highly predictable template: company name, ticker, stock movement, timing, earnings figures, year-over-year comparisons, and analyst estimates. Nearly every word is highly probable given its context, with most log probabilities close to zero. The model easily predicts phrases like "shares fell," "the company reported," "compared to," and "analysts surveyed by" because they constitute standard financial news boilerplate. The few words with lower probabilities (e.g., "software-as-a-service," "Adjusted") reflect industry-specific terminology rather than narrative unpredictability.

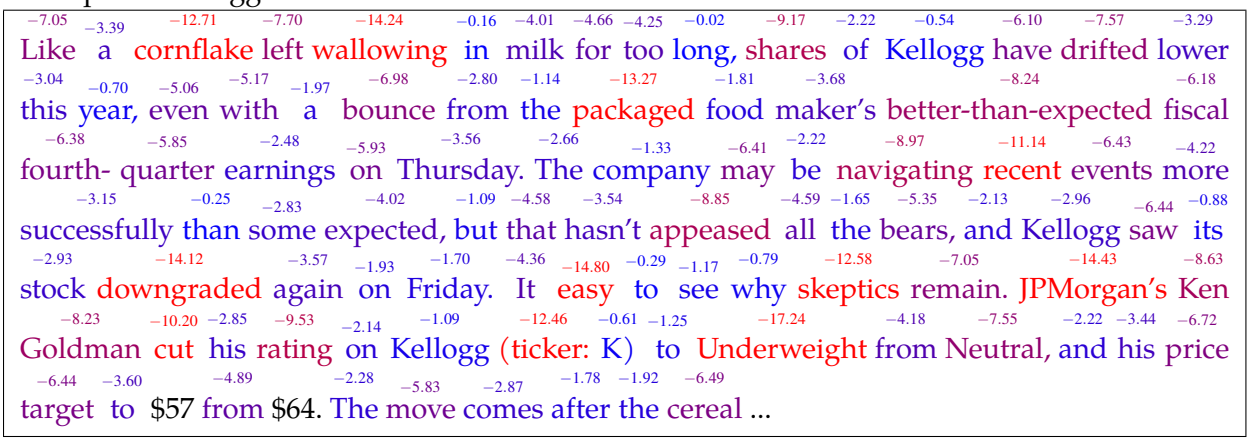
Example 2: Amgen




The Amgen article similarly demonstrates low perplexity through its adherence to standard earnings announcement conventions. The language model assigns high probability to the recurring syntactic patterns: "shares rose more than," "the company reported," and the formulaic earnings-per-share comparisons. Even longer phrases like "compared with... billion, or... a share, in the year-ago period" are highly predictable because this exact structure appears ubiquitously in financial reporting. The text provides no cognitive jolt—an investor encountering this article processes it through well-worn mental templates.

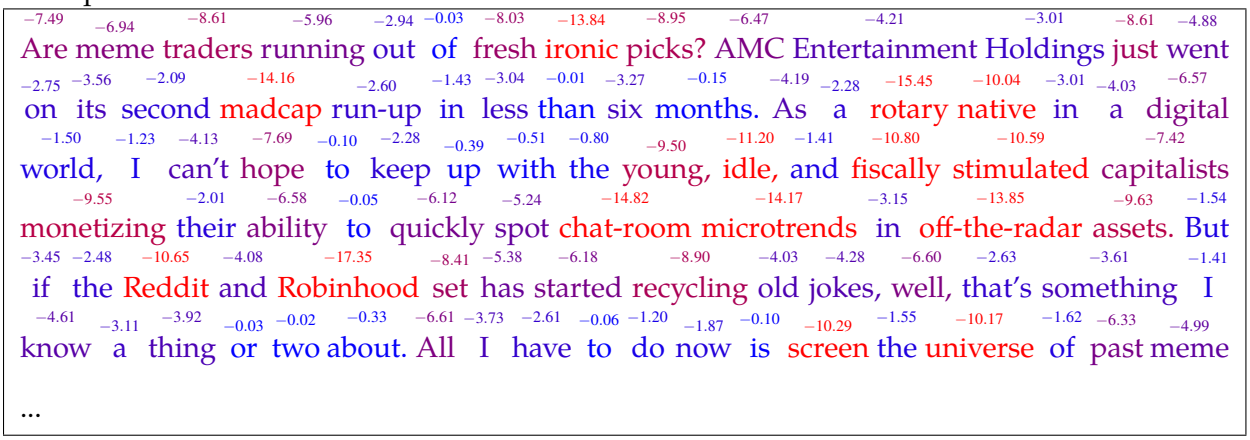
## High Perplexity Examples

Example 3: Kellogg's



The Kellogg's article demonstrates high perplexity through its striking opening metaphor: "Like a cornflake left wallowing in milk for too long." The model assigns extremely low probability to "cornflake" and "wallowing" in this context because financial news seldom employs such vivid, product-related imagery. The metaphor uses accessible, everyday language—yet it is highly unpredictable in a financial context. This exemplifies the distinction between syntactic complexity and narrative unpredictability. The creative framing forces readers to pause and process, potentially triggering the deeper engagement that my hypothesis predicts.

Example 4: AMC Entertainment




The AMC Entertainment article represents perhaps the most striking contrast with conventional financial journalism. The opening rhetorical question ("Are meme traders running out of fresh ironic picks?") immediately signals an unconventional narrative approach. The author's self-deprecating characterization as a "rotary native in a digital world" is deeply unpredictable in financial coverage—the model assigns log probability of −15.45 to "rotary." The colorful description of retail traders as "young, idle, and fiscally stimulated capitalists monetizing their ability to quickly spot chat-room microtrends in off-the-radar assets" packs multiple high-surprise words into a single phrase. Terms like "madcap," "chat-room," "microtrends," "off-the-radar," "Reddit," and "Robinhood" each violate the model's expectations about financial news language. This article represents financial commentary filtered through a distinctly personal, humorous voice—a sharp

departure from the detached, formulaic reporting of the low perplexity examples.

These examples illustrate how perplexity captures narrative unpredictability rather than mere complexity. The high perplexity articles use informal, colloquial language composed of simple words. Yet the *combination* and *context* of these words—metaphors comparing stocks to soggy cereal, self-aware commentary about meme trading—creates cognitive surprise that the perplexity measure detects. This is precisely the type of text that, according to the dual-process framework, should disrupt automatic processing and trigger deeper analytical engagement.

### 2.1.4 Implementation and Measurement

I make several methodological choices in implementing this measure. First, while LLMs operate at the token level, humans process text at the word level (Rayner, 1998). For words spanning multiple tokens, I consolidate token-level log probabilities to the word level before applying Equation 1. I also exclude non-alphabetic tokens, such as punctuation and numbers, to focus purely on narrative style.

Second, perplexity is calculated for a fixed portion of each article. Readers engage in selective information acquisition and are heavily influenced by information presented first—the primacy effect (Asch, 1946). Accordingly, I include only the first 100 words of each article. This aligns with evidence from financial disclosures showing that the market reacts most strongly to initial sections (Davis et al., 2012) and that investors rarely read documents completely (Dyer et al., 2017). This focus on opening text captures the content most likely to deliver the initial cognitive jolt.

Third, to isolate narrative unpredictability from entity-name rarity, I prepend each article's context window ($\{w_1, \ldots, w_{i-1}\}$ in Equation 1) with a standardized preface: "News article about {company} ({ticker}):". This conditions the model on the article's subject, ensuring that high perplexity reflects unconventional narrative rather than sur-

prise at uncommon company names. The preface tokens are excluded from the perplexity calculation.

For firms with multiple articles on a single day, I select the article with the highest perplexity score. This choice is theory-driven: my hypothesis posits that high perplexity triggers System 2 processing. By selecting the maximum-perplexity article, I test the effect of the most potent strain-inducing stimulus available to investors, analogous to focusing on the most salient signal (Bordalo, Gennaioli, and Shleifer, 2012).

#### 2.1.5 Temporal Alignment with ChronoGPT

To construct a robust perplexity measure, I must ensure that the language model reflects the state of linguistic and financial knowledge available to investors at the time of the article's release, thus avoiding lookahead bias. Standard LLMs like GPT-4 are trained on internet corpora including future information—financial outcomes, subsequent news coverage, even academic papers discussing the events being measured. Using such models would contaminate perplexity scores with information unavailable to contemporaneous investors.

I address this challenge using ChronoGPT, a set of time-stamped large language models developed by He, Lv, Manela, and Wu (2025). These chronologically consistent models are trained only on text data available at each specific point in time. Despite this strict temporal constraint, ChronoGPT achieves strong performance on natural language processing benchmarks, demonstrating sophisticated understanding of language and financial norms for any given historical period.

Specifically, for a news article released in year $t$, I use ChronoGPT$_{t-1}$ to evaluate its perplexity. This design ensures that the model is familiar with linguistic norms through the prior year but remains unexposed to the current article and subsequent events and language. This temporal alignment prevents false familiarity, where the model

might otherwise assign high probability (low perplexity) to a word or concept that was genuinely novel to investors at publication. The resulting perplexity score serves as my primary, time-aligned independent variable.

## 2.2 Financial News Data

I utilize the Dow Jones Newswire dataset, a real-time newswire to which major institutional investors subscribe. This dataset provides extensive coverage of financial markets, economies, and companies, aggregating reports from leading media sources such as The Wall Street Journal, Barron's, and MarketWatch.

The dataset is highly granular, including news headlines, full article texts, and precise display timestamps that record the time when the news becomes available to clients. To ensure firm-specific relevance and avoid confounding effects, the sample is restricted to articles that are reliably tagged to exactly one ticker. I focus my analysis exclusively on firm-specific news with over 100 words that can be reliably attributed to a single company, following Ke, Kelly, and Xiu (2019).

My dataset covers January 2008 to July 2023. Articles are mapped to trading days from the close of the previous trading day $(t-1)$ to the close of the current trading day $(t)$. Articles published after market close are mapped to the next trading day $(t+1)$.

### 2.2.1 Isolating High-Impact and Directional News

My hypothesis centers on underreaction to news. To test this, I construct a sample meeting two criteria: (i) economically significant news events and (ii) clear, identifiable directional impact ("good" or "bad" news).

First, I isolate high-salience, non-discretionary news events using Dow Jones subject tags, excluding company-issued press releases and non-8-K filings. These exclusions

are justified because statutory filings adhere to rigid legal formats that obscure stylistic variation, while press releases are often boilerplate or promotional (Davis, Piger, and Sedor, 2012). I focus on articles tagged as Corporate Actions, comprising 64.8% of the remaining articles. This category includes events known to have significant market impact: Earnings (24.5%), Personnel Changes (19.3%), and Mergers and Acquisitions (11.5%) (Andrade, Mitchell, and Stafford, 2001; Ball and Brown, 1968; Li, 2010).

Second, I identify each article's direction and impact using a text-based measure of predicted total returns, applying the methodology of Chen, Kelly, and Xiu (2022). While their original application predicts next-day returns, I adapt their framework to predict cumulative 40-day returns to capture total, fundamental news impact—the benchmark against which underreaction is measured.

For this task, I use ChronoBERT, time-stamped auto-encoding (BERT-style) models from He, Lv, Manela, and Wu (2025). Each article is analyzed using a model trained only on data through year $t-1$, ensuring return predictions are free of lookahead bias. An additional year (2007) serves as a "burn-in" period for initial model estimation.

I transform each article's text into ChronoBERT embeddings ($e_{it}$), link each to the stock's 40-day cumulative return ($r_{i[t,t+40]}$), and fit monthly Fama-MacBeth regressions with ridge penalties:

$$\min_{\alpha_m,\beta_m} \sum_{i=1}^{N_m} (r_{i[t,t+40]} - \alpha_m - \beta_m' e_{it})^2 + \lambda ||\beta_m||_2^2, \tag{2}$$

where $N_m$ is the number of news articles in month $m$, $e_{it}$ represents the news article embeddings, $||\beta_m||_2^2$ is the squared L2-norm of the coefficient vector, and $\lambda$ is the ridge penalty parameter, determined via leave-one-out cross-validation from grid points ranging from $10^{-10}$ to $10^{10}$.

To construct real-time out-of-sample forecasts, in each month $m'$, I use an average of

the estimated coefficients from all previous months' cross-sectional models:

$$\hat{r}_{i[t,t+40]} = \bar{\alpha}_{m'} + \bar{\beta}'_{m'} e_{i,t}, \tag{3}$$

where $\bar{\alpha}_{m'} = \frac{1}{m'-1}\sum_{m=1}^{m'-1}\hat{\alpha}_m$ and $\bar{\beta}_{m'} = \frac{1}{m'-1}\sum_{m=1}^{m'-1}\hat{\beta}_m$.

Using these predictions, I sort stocks daily and retain the top and bottom 33% of articles, yielding a final sample of 129,334 stock-days with strong "Positive News" and "Negative News" signals.

The use of ChronoBERT for filtering while using ChronoGPT for perplexity analysis is deliberate. BERT-style (auto-encoding) models are designed for predictive tasks generating rich contextual embeddings. GPT-style (auto-regressive) models are designed for calculating sequential probabilities (perplexity). By using two different, state-of-the-art model architectures for my two key tasks, I ensure that my main independent variable (ChronoGPT-derived perplexity) is not mechanically correlated with my sample selection filter (ChronoBERT-derived predicted returns). This methodological separation is critical to avoid confounding results that could arise from using the same model's internal logic for both measurement and filtering.

## 2.3 Stock Returns Data

Stock returns data are obtained from CRSP, including U.S. common stocks with share codes 10 and 11. To estimate abnormal returns, I employ the Fama-French Four-Factor Model (Fama and French, 1993; Carhart, 1997):

$$R_{it} - R_t^f = \alpha_i + \beta_i^{MKT}(R_t^{MKT} - R_t^f) + \beta_i^{SMB} SMB_t + \beta_i^{HML} HML_t + \beta_i^{MOM} MOM_t + \varepsilon_{i,t}. \tag{4}$$

Factor sensitivities are estimated using a one-year (252 trading day) rolling window concluding 21 trading days prior to the event window. This ensures factor loadings are

determined solely by historical data, mitigating lookahead bias.

# 3 Results

## 3.1 News Perplexity and Underreaction: Initial Evidence

My central empirical test investigates whether market underreaction—the well-documented phenomenon where prices drift gradually in the direction of initial news shocks (Ball and Brown, 1968; Bernard and Thomas, 1990; Chan et al., 1996)—varies with news perplexity. If my hypothesis holds, higher perplexity news should trigger deeper System 2 processing, leading to faster, more complete information incorporation and thus less underreaction.

Table 1 summarizes average abnormal returns for each perplexity level—Low, Moderate, and High—where "Low Perplexity" corresponds to fluent, predictable text and "High Perplexity" corresponds to strain-inducing text.

Panel A presents results for positive news. Column (1) shows that the initial response (abnormal returns at $t = [0,1]$) increases monotonically with perplexity from 0.50% to 1.87%. Column (2) shows that the delayed response (abnormal returns at $t = [2,40]$) decreases monotonically from 0.83% to 0.04%. Column (3) shows total response (abnormal returns at $t = [0,40]$) varying between 0.96% and 1.90%. Column (4) presents the underreaction ratio—delayed response as a fraction of total response—which decreases monotonically from 0.62 for low perplexity to 0.02 for high perplexity. The difference of 0.60 is highly significant. Standard errors are double-clustered at stock and day level, and those of the underreaction ratio are estimated via the delta method.

Panel B presents parallel results for negative news. Initial response increases from -0.63% to -2.49%; delayed response decreases from -0.62% to -0.17%; the underreaction ratio falls from 0.50 to 0.06, with the difference of 0.43 again highly significant. These

## **Table 1** Underreaction Decreases with Perplexity

This table presents initial evidence examining the effect of news perplexity on the market's underreaction to news. The sample consists of 129,334 high-impact, firm-specific Corporate Action news events from the Dow Jones Newswire (2008-2023), split into positive news (Panel A, N=65,312) and negative news (Panel B, N=64,022), which represent the top and bottom tertiles identified based on predicted two-month returns using text embeddings of news articles, following the methodology of Chen, Kelly, and Xiu (2022). The table reports the initial response ($ARet^{[0,1]}$) in Column (1), delayed response ($ARet^{[2,40]}$) in Column (2), and total response ($ARet^{[0,40]}$) in Column (3) for low perplexity (bottom decile), moderate perplexity (middle 80%), and high perplexity (top decile) news articles. Abnormal returns are calculated using the Fama-French four-factor model plus momentum, estimated using a one-year (252 trading day) rolling window of daily returns, concluding 21 trading days prior to the first event day. Standard errors are two-way clustered by firm and day. In Column (4), the Underreaction Ratio is the average delayed response as a fraction of average total response. Standard errors for the Underreaction Ratio and the difference between low and high perplexity groups (Low − High) are calculated using the Delta method and reported in parentheses.

**Panel A: Positive News**

| | **Initial Response** $(ARet^{[0,1]})$ (1) | **Delayed Response** $(ARet^{[2,40]})$ (2) | **Total Response** $(ARet^{[0,40]})$ (3) | **Underreaction Ratio** $(E[ARet^{[2,40]}]/E[ARet^{[0,40]}])$ $(4) = \frac{(2)}{(3)}$ |
|---|---|---|---|---|
| Low Perplexity | 0.50 | 0.83 | 1.33 | 0.62 |
| | (0.11) | (0.23) | (0.25) | (0.08) |
| Moderate Perplexity | 0.91 | 0.05 | 0.96 | 0.05 |
| | (0.05) | (0.13) | (0.14) | (0.12) |
| High Perplexity | 1.87 | 0.04 | 1.90 | 0.02 |
| | (0.16) | (0.27) | (0.31) | (0.14) |
| Low − High | | | | 0.60*** |
| | | | | (0.16) |
| Observations | 65,312 | 65,312 | 65,312 | 65,312 |
| Adjusted $R^2$ | 0.00101 | 0.00008 | 0.00012 | |

**Panel B: Negative News**

| | **Initial Response** $(ARet^{[0,1]})$ (1) | **Delayed Response** $(ARet^{[2,40]})$ (2) | **Total Response** $(ARet^{[0,40]})$ (3) | **Underreaction Ratio** $(E[ARet^{[2,40]}]/E[ARet^{[0,40]}])$ $(4) = \frac{(2)}{(3)}$ |
|---|---|---|---|---|
| Low Perplexity | -0.63 | -0.62 | -1.25 | 0.50 |
| | (0.09) | (0.30) | (0.32) | (0.13) |
| Moderate Perplexity | -0.85 | -0.21 | -1.06 | 0.20 |
| | (0.05) | (0.14) | (0.16) | (0.10) |
| High Perplexity | -2.49 | -0.17 | -2.66 | 0.06 |
| | (0.18) | (0.25) | (0.30) | (0.09) |
| Low − High | | | | 0.43*** |
| | | | | (0.15) |
| Observations | 64,022 | 64,022 | 64,022 | 64,022 |
| Adjusted $R^2$ | 0.00271 | 0.00001 | 0.00037 | |

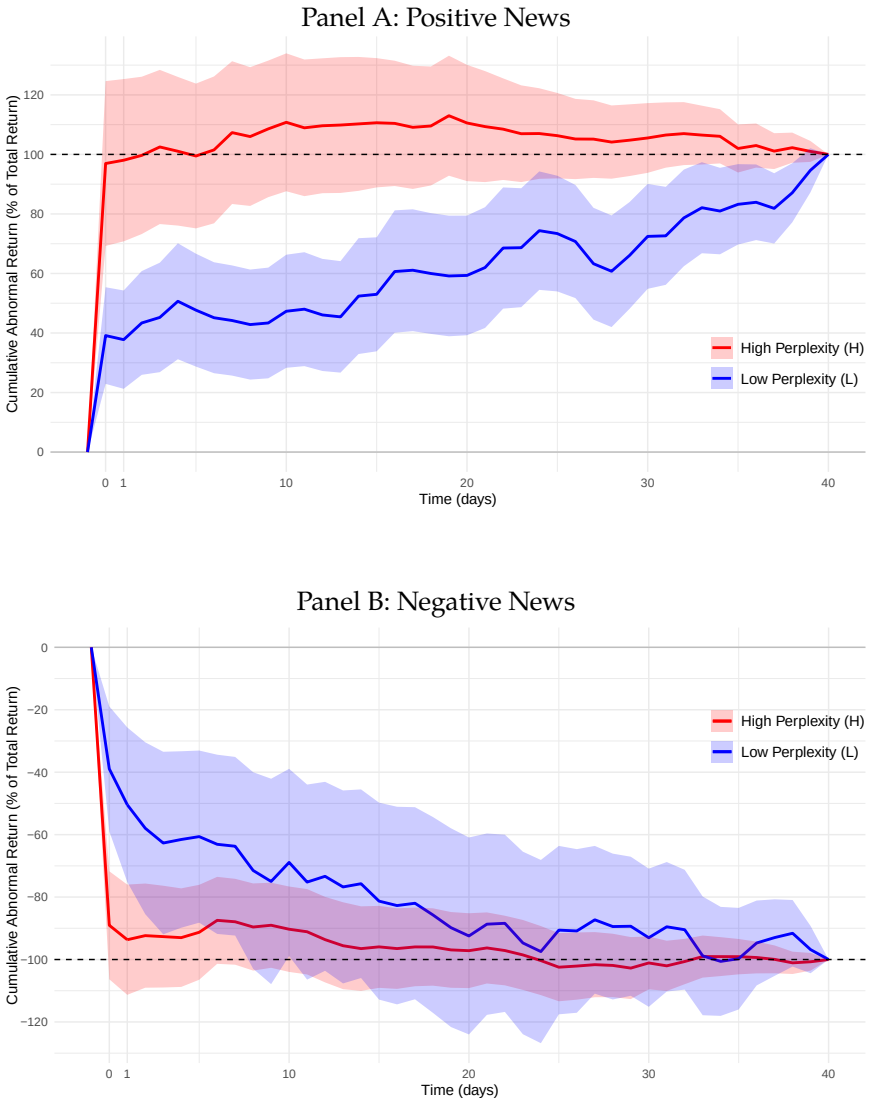



**Figure 1** Cumulative Abnormal Returns Following News Event

This figure illustrates the time path of price adjustment, showing the Cumulative Abnormal Return (CAR) as a ratio of the total reaction over the 40-day event window, showing positive news (Panel A) and negative news (Panel B) separately, which represent the top and bottom tertiles of Corporate Action news identified based on predicted two-month returns using text embeddings of news articles, following the methodology of Chen, Kelly, and Xiu (2022). Low perplexity corresponds to the bottom decile and high perplexity corresponds to the top decile of perplexity scores on each day. The vertical axis plots the cumulative abnormal return ratio, calculated as $CAR^{[0,h]} = \frac{E[ARet^{[0,h]}]}{E[ARet^{[0,40]}]}$, where abnormal returns are calculated using the Fama-French four-factor model plus momentum, estimated using a one-year (252 trading day) rolling window of daily returns, concluding 21 trading days prior to the first event day. The blue line represents the low perplexity group and the red line represents the high perplexity group. The shaded area represents 95% confidence intervals, with standard errors two-way clustered by firm and day.

results suggest that when text is cognitively demanding, investors process information more completely upfront, leaving little to be incorporated later.

Figure 1 provides visual representation. It plots the cumulative abnormal return ratio, $CAR^{[0,h]} = \frac{E[ARet^{[0,h]}]}{E[ARet^{[0,40]}]}$, over the 40-day event window. For both positive and negative news, the high perplexity line (red) jumps almost immediately to 100% around day 0 or 1 and remains flat—price adjustment is essentially complete within two days when text induces strain. The low perplexity line (blue) shows a smaller initial jump followed by gradual drift over subsequent weeks as the market inefficiently processes the predictable narrative.

## 3.2 News Perplexity and Underreaction: Main Results

To provide a more unified and robust test, I follow Dellavigna and Pollet (2009) and examine the Delayed Response Ratio ($DRR$). This approach has two advantages. First, while the underreaction ratio in Table 1 measures reaction completeness within a single news direction, the DRR provides a summary statistic for how completely the market distinguishes between good and bad news in the initial window versus the delayed window. My hypothesis predicts that high perplexity should lead to more complete initial incorporation and lower DRR. Second, this method allows pooling positive and negative news into a single, more powerful regression framework.

Following Dellavigna and Pollet (2009), I define the Delayed Response Ratio ($DRR$) as the ratio of the differential delayed response to the differential long-term response:

$$DRR = \frac{E\left[ARet^{[2,40]} \,\middle|\, pos\right] - E\left[ARet^{[2,40]} \,\middle|\, neg\right]}{E\left[ARet^{[0,40]} \,\middle|\, pos\right] - E\left[ARet^{[0,40]} \,\middle|\, neg\right]} \tag{5}$$

To estimate DRR components, I run the following pooled OLS regression comparing

stock responses to positive news ($I_{it}^{pos} = 1$) versus negative news ($I_{it}^{pos} = 0$):

$$\begin{aligned} ARet_{it}^{[h,H]} = {} & \phi_{pos-neg} I_{it}^{pos} + \phi^{Moderate} I_{it}^{Moderate} + \phi^{High} I_{it}^{High} \\ & + \phi_{pos-neg}^{Moderate} I_{it}^{pos} I_{it}^{Moderate} + \phi_{pos-neg}^{High} I_{it}^{pos} I_{it}^{High} \\ & + \Gamma_{neg} \mathbf{X}_{it} + \Gamma_{pos-neg} I_{it}^{pos} \mathbf{X}_{it} + \varepsilon_{it}, \end{aligned} \tag{6}$$

where $ARet_{it}^{[h,H]}$ denotes the Fama-French four-factor abnormal stock return for company $i$ between days $h$ and $H$ following the news publication. I estimate this regression separately for three horizons: initial response [0,1], delayed response [2,40], and total response [0,40]. The DRR is then computed using the ratio of coefficients from the delayed and total regressions. The sample includes only positive news and negative news. The base group is negative news with low perplexity. $I_{it}^{Moderate}$ and $I_{it}^{High}$ are indicator variables for moderate and high perplexity. $\mathbf{X}_{it}$ denotes a vector of control variables, also interacted with the positive news indicator.

Continuous control variables in $X_{it}$ are demeaned separately within the positive and negative subsamples. This group-wise demeaning ensures that the main effect coefficients ($\phi$) capture the average return levels of the representative stock in each group, preserving the magnitude of the return spread used in the DRR denominator. These controls include centered day-of-the-week fixed effects, stock characteristics (volatility, illiquidity (Amihud, 2002), market cap, volume of news coverage), content (predicted total long-term return from LLM embeddings of the article, following Chen, Kelly, and Xiu, 2022), syntax (article length, Fog-index), and news subject fixed effects (earnings, personnel, M&A).

I focus on coefficients $\phi_{pos-neg}$ (differential return for low perplexity) and $\phi_{pos-neg}^{High}$ (additional differential reaction for high perplexity). If high perplexity accelerates information incorporation, $\phi_{pos-neg}^{High}$ should be positive for initial response (larger immediate divergence) and negative for delayed response (less subsequent drift).

## **Table 2** Delayed Response Ratio Decreases with Perplexity

This table presents results from a pooled OLS regression examining the effect of news perplexity on the market's differential response to positive news versus negative news. Low perplexity corresponds to the bottom decile, moderate perplexity to the middle 80%, and high perplexity to the top decile of perplexity scores on each day. The dependent variable is the Cumulative Abnormal Return ($ARet$) over the specified window. The regression specification is detailed in Equation 6. Panels A, B, and C report coefficients for the initial ($ARet^{[0,1]}$), delayed ($ARet^{[2,40]}$), and total ($ARet^{[0,40]}$) responses, respectively. Panel D reports the implied Delayed Response Ratio ($DRR$) calculated using Panels B and C via Equations 7 and 8, and reports the results of a one-sided test evaluating whether the difference between ratios is significantly greater than zero. Continuous control variables in $X_{it}$ are demeaned separately within the positive and negative subsamples. Control sets are added incrementally and are interacted with $I^{pos}$: centered day-of-the-week fixed effects, stock characteristics (volatility, illiquidity, market cap, volume of news coverage), content (predicted two-month returns using text embeddings), syntax (article length, Fog-index), and news subject fixed effects (earnings, personnel, M&A). Standard errors (in parentheses) are two-way clustered by firm and day, and additionally, standard errors for Panel D use the Delta method. In Panels A, B and C, asterisks (***, **, *) indicate significance at the 1%, 5%, and 10% levels based on two-sided p-values. In Panel D, asterisks indicate significance at the same levels based on one-sided p-values for the difference in ratios.

**Panel A: Initial Response** ($ARet^{[0,1]}$)

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| Positive News ($\phi_{pos-neg}$) | 1.13*** | 1.04*** | 1.06*** | 1.10*** | 1.61*** |
| | (0.14) | (0.14) | (0.15) | (0.14) | (0.15) |
| Moderate Perplexity ($\phi^{Moderate}$) | -0.22** | -0.26*** | -0.26*** | -0.20** | 0.04 |
| | (0.10) | (0.10) | (0.10) | (0.10) | (0.10) |
| High Perplexity ($\phi^{High}$) | -1.86*** | -1.94*** | -1.92*** | -1.59*** | -0.95*** |
| | (0.20) | (0.20) | (0.20) | (0.20) | (0.20) |
| Positive News × Moderate Perplexity ($\phi^{Moderate}_{pos-neg}$) | 0.63*** | 0.56*** | 0.55*** | 0.55*** | 0.09 |
| | (0.15) | (0.15) | (0.15) | (0.15) | (0.15) |
| Positive News × High Perplexity ($\phi^{High}_{pos-neg}$) | 3.23*** | 3.33*** | 3.23*** | 2.91*** | 1.74*** |
| | (0.28) | (0.28) | (0.28) | (0.28) | (0.28) |
| Positive News × Stock Controls | | Yes | Yes | Yes | Yes |
| Positive News × Industry FE | | | Yes | Yes | Yes |
| Positive News × Day-of-the-Week FE | | | Yes | Yes | Yes |
| Positive News × Content | | | | Yes | Yes |
| Positive News × Syntax Controls | | | | Yes | Yes |
| Positive News × News Subject FE | | | | | Yes |
| Observations | 129,334 | 129,334 | 129,334 | 129,334 | 129,334 |
| Adjusted $R^2$ | 0.01140 | 0.01931 | 0.02040 | 0.02618 | 0.03167 |

**Panel B: Delayed Response** ($ARet^{[2,40]}$)

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| Positive News ($\phi_{pos-neg}$) | 1.44*** | 1.26*** | 1.24*** | 1.23*** | 1.06*** |
| | (0.37) | (0.37) | (0.37) | (0.37) | (0.37) |
| Moderate Perplexity ($\phi^{Moderate}$) | 0.40 | 0.46 | 0.45 | 0.47 | 0.40 |
| | (0.32) | (0.32) | (0.32) | (0.32) | (0.32) |
| High Perplexity ($\phi^{High}$) | 0.45 | 0.57 | 0.48 | 0.54 | 0.30 |
| | (0.39) | (0.39) | (0.39) | (0.39) | (0.41) |
| Positive News × Moderate Perplexity ($\phi^{Moderate}_{pos-neg}$) | -1.18*** | -1.07*** | -1.07*** | -1.06*** | -0.89** |
| | (0.39) | (0.39) | (0.39) | (0.39) | (0.40) |
| Positive News × High Perplexity ($\phi^{High}_{pos-neg}$) | -1.24** | -1.04* | -0.97* | -0.99* | -0.64 |
| | (0.50) | (0.53) | (0.54) | (0.54) | (0.56) |
| Positive News × Stock Controls | | Yes | Yes | Yes | Yes |
| Positive News × Industry FE | | | Yes | Yes | Yes |
| Positive News × Day-of-the-Week FE | | | Yes | Yes | Yes |
| Positive News × Content | | | | Yes | Yes |
| Positive News × Syntax Controls | | | | Yes | Yes |
| Positive News × News Subject FE | | | | | Yes |
| Observations | 129,334 | 129,334 | 129,334 | 129,334 | 129,334 |
| Adjusted $R^2$ | 0.00012 | 0.00183 | 0.00261 | 0.00262 | 0.00284 |

**Panel C: Total Response** ($ARet^{[0,40]}$)

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| Positive News ($\phi_{pos-neg}$) | 2.57*** | 2.30*** | 2.31*** | 2.34*** | 2.67*** |
| | (0.39) | (0.39) | (0.40) | (0.39) | (0.40) |
| Moderate Perplexity ($\phi^{Moderate}$) | 0.18 | 0.21 | 0.19 | 0.28 | 0.44 |
| | (0.34) | (0.34) | (0.34) | (0.34) | (0.34) |
| High Perplexity ($\phi^{High}$) | -1.41*** | -1.37*** | -1.44*** | -1.05** | -0.64 |
| | (0.44) | (0.44) | (0.45) | (0.45) | (0.47) |
| Positive News × Moderate Perplexity ($\phi^{Moderate}_{pos-neg}$) | -0.54 | -0.51 | -0.51 | -0.51 | -0.81* |
| | (0.43) | (0.43) | (0.43) | (0.43) | (0.43) |
| Positive News × High Perplexity ($\phi^{High}_{pos-neg}$) | 1.99*** | 2.28*** | 2.26*** | 1.92*** | 1.10* |
| | (0.58) | (0.61) | (0.61) | (0.61) | (0.63) |
| Positive News × Stock Controls | | Yes | Yes | Yes | Yes |
| Positive News × Industry FE | | | Yes | Yes | Yes |
| Positive News × Day-of-the-Week FE | | | Yes | Yes | Yes |
| Positive News × Content | | | | Yes | Yes |
| Positive News × Syntax Controls | | | | Yes | Yes |
| Positive News × News Subject FE | | | | | Yes |
| Observations | 129,334 | 129,334 | 129,334 | 129,334 | 129,334 |
| Adjusted $R^2$ | 0.00277 | 0.00545 | 0.00614 | 0.00750 | 0.00821 |

**Panel D: Delayed Response Ratio**

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| Delayed Response Ratio for Low Perplexity ($DRR^{Low}$) | 0.56 | 0.55 | 0.54 | 0.53 | 0.40 |
| | (0.07) | (0.08) | (0.08) | (0.08) | (0.09) |
| Delayed Response Ratio for High Perplexity ($DRR^{High}$) | 0.05 | 0.05 | 0.06 | 0.06 | 0.11 |
| | (0.07) | (0.09) | (0.09) | (0.09) | (0.10) |
| $DRR^{Low} - DRR^{High}$ | 0.52*** | 0.50*** | 0.48*** | 0.47*** | 0.29** |
| | (0.10) | (0.11) | (0.11) | (0.12) | (0.13) |
| Positive News × Stock Controls | | Yes | Yes | Yes | Yes |
| Positive News × Industry FE | | | Yes | Yes | Yes |
| Positive News × Day-of-the-Week FE | | | Yes | Yes | Yes |
| Positive News × Content | | | | Yes | Yes |
| Positive News × Syntax Controls | | | | Yes | Yes |
| Positive News × News Subject FE | | | | | Yes |
| Observations | 129,334 | 129,334 | 129,334 | 129,334 | 129,334 |

From estimated coefficients, the DRR for low and high perplexity groups is:

$$DRR^{Low} = \frac{\phi_{pos-neg}^{[2,40]}}{\phi_{pos-neg}^{[0,40]}}, \tag{7}$$

$$DRR^{High} = \frac{\phi_{pos-neg}^{[2,40]} + \phi_{pos-neg}^{High[2,40]}}{\phi_{pos-neg}^{[0,40]} + \phi_{pos-neg}^{High[0,40]}}. \tag{8}$$

Table 2 presents the results. Panels A, B, and C present the regression results for the initial response ($ARet^{[0,1]}$), delayed response ($ARet^{[2,40]}$), and total response ($ARet^{[0,40]}$), respectively. Panel D presents the derived DRR.

Panel A (Initial Response) shows that for low perplexity news, the market differentiates between positive and negative news by 1.13% ($\phi_{pos-neg}$). For high perplexity news, this initial differentiation is significantly larger, increasing by 3.23% ($\phi_{pos-neg}^{High}$). This suggests cognitively demanding text triggers a more immediate and powerful market reaction. The effect remains robust across specifications adding controls progressively; in the full specification (Column 5), the differential initial response for high perplexity remains large (1.74%) and highly significant.

Panel B (Delayed Response) shows the pattern reverses, consistent with the "wake-up call" hypothesis. For low perplexity news, substantial delayed divergence (1.06%–1.26%) exists between positive and negative news. For high perplexity news, delayed divergence is smaller by 0.64%–1.24%.

Panel D presents the core results: the Delayed Response Ratio. In Column (1), $DRR^{Low}$ is 0.56, meaning 56% of total market reaction to low perplexity news occurs with delay. $DRR^{High}$ is just 0.05. The difference of 0.52 is highly significant. In the full specification (Column 5), $DRR^{Low}$ is 0.40 and $DRR^{High}$ is 0.11. The difference remains positive and significant at 0.29.

These results provide strong support for the central hypothesis. High perplexity

## **Table 3** Delayed Response Ratio Decreases with Perplexity: Large vs Small Firms

This table replicates the analysis in Table 2, with subsamples separated by the median firm size of all articles on each day. Low perplexity corresponds to the bottom decile, moderate perplexity to the middle 80%, and high perplexity to the top decile of perplexity scores on each day.

**Panel A: Initial Response** ($ARet^{[0,1]}$)

| | Large | | Small | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Positive News ($\phi_{pos-neg}$) | 0.72*** | 1.05*** | 1.73*** | 2.31*** |
| | (0.12) | (0.13) | (0.23) | (0.24) |
| Moderate Perplexity ($\phi^{Moderate}$) | -0.20** | 0.01 | -0.29 | 0.00 |
| | (0.08) | (0.09) | (0.18) | (0.18) |
| High Perplexity ($\phi^{High}$) | -1.30*** | -0.70*** | -2.59*** | -1.39*** |
| | (0.16) | (0.16) | (0.33) | (0.34) |
| Positive News × Moderate Perplexity ($\phi^{Moderate}_{pos-neg}$) | 0.32** | 0.02 | 0.56** | 0.05 |
| | (0.13) | (0.13) | (0.24) | (0.24) |
| Positive News × High Perplexity ($\phi^{High}_{pos-neg}$) | 2.11*** | 1.24*** | 4.63*** | 2.81*** |
| | (0.22) | (0.21) | (0.47) | (0.47) |
| Positive News × Stock Controls | Yes | Yes | Yes | Yes |
| Positive News × Industry FE | Yes | Yes | Yes | Yes |
| Positive News × Day-of-the-Week FE | Yes | Yes | Yes | Yes |
| Positive News × Content | | Yes | | Yes |
| Positive News × Syntax Controls | | Yes | | Yes |
| Positive News × News Subject FE | | Yes | | Yes |
| Observations | 63,626 | 63,626 | 65,708 | 65,708 |
| Adjusted $R^2$ | 0.02431 | 0.03672 | 0.02242 | 0.03736 |

**Panel B: Delayed Response** ($ARet^{[2,40]}$)

| | Large | | Small | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Positive News ($\phi_{pos-neg}$) | 0.48 | 0.44 | 1.76*** | 1.49** |
| | (0.30) | (0.31) | (0.59) | (0.60) |
| Moderate Perplexity ($\phi^{Moderate}$) | 0.33 | 0.30 | 1.04** | 1.05** |
| | (0.23) | (0.23) | (0.51) | (0.52) |
| High Perplexity ($\phi^{High}$) | -0.04 | -0.03 | 1.23* | 0.95 |
| | (0.31) | (0.31) | (0.74) | (0.78) |
| Positive News × Moderate Perplexity ($\phi^{Moderate}_{pos-neg}$) | -0.54* | -0.49 | -1.31** | -1.08* |
| | (0.30) | (0.31) | (0.64) | (0.65) |
| Positive News × High Perplexity ($\phi^{High}_{pos-neg}$) | -0.58 | -0.57 | -1.95** | -1.34 |
| | (0.46) | (0.46) | (0.99) | (1.04) |
| Positive News × Stock Controls | Yes | Yes | Yes | Yes |
| Positive News × Industry FE | Yes | Yes | Yes | Yes |
| Positive News × Day-of-the-Week FE | Yes | Yes | Yes | Yes |
| Positive News × Content | | Yes | | Yes |
| Positive News × Syntax Controls | | Yes | | Yes |
| Positive News × News Subject FE | | Yes | | Yes |
| Observations | 63,626 | 63,626 | 65,708 | 65,708 |
| Adjusted $R^2$ | 0.00492 | 0.00506 | 0.00319 | 0.00363 |

**Panel C: Total Response** ($ARet^{[0,40]}$)

| | Large | | Small | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Positive News ($\phi_{pos-neg}$) | 1.20*** | 1.48*** | 3.49*** | 3.80*** |
| | (0.32) | (0.32) | (0.63) | (0.65) |
| Moderate Perplexity ($\phi^{Moderate}$) | 0.13 | 0.31 | 0.75 | 1.05* |
| | (0.24) | (0.24) | (0.54) | (0.55) |
| High Perplexity ($\phi^{High}$) | -1.34*** | -0.73** | -1.36* | -0.44 |
| | (0.34) | (0.35) | (0.82) | (0.86) |
| Positive News × Moderate Perplexity ($\phi^{Moderate}_{pos-neg}$) | -0.22 | -0.47 | -0.76 | -1.02 |
| | (0.33) | (0.33) | (0.68) | (0.69) |
| Positive News × High Perplexity ($\phi^{High}_{pos-neg}$) | 1.53*** | 0.66 | 2.68** | 1.47 |
| | (0.50) | (0.50) | (1.10) | (1.15) |
| Positive News × Stock Controls | Yes | Yes | Yes | Yes |
| Positive News × Industry FE | Yes | Yes | Yes | Yes |
| Positive News × Day-of-the-Week FE | Yes | Yes | Yes | Yes |
| Positive News × Content | | Yes | | Yes |
| Positive News × Syntax Controls | | Yes | | Yes |
| Positive News × News Subject FE | | Yes | | Yes |
| Observations | 63,626 | 63,626 | 65,708 | 65,708 |
| Adjusted $R^2$ | 0.00739 | 0.01009 | 0.00708 | 0.00992 |

**Panel D: Delayed Response Ratio**

| | Large | | Small | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Delayed Response Ratio for Low Perplexity ($DRR^{Low}$) | 0.40 | 0.30 | 0.51 | 0.39 |
| | (0.16) | (0.15) | (0.09) | (0.14) |
| Delayed Response Ratio for High Perplexity ($DRR^{High}$) | -0.03 | -0.06 | -0.03 | 0.03 |
| | (0.14) | (0.18) | (0.14) | (0.16) |
| $DRR^{Low} - DRR^{High}$ | 0.44** | 0.36* | 0.53*** | 0.36** |
| | (0.20) | (0.22) | (0.16) | (0.19) |
| Positive News × Stock Controls | Yes | Yes | Yes | Yes |
| Positive News × Industry FE | Yes | Yes | Yes | Yes |
| Positive News × Day-of-the-Week FE | Yes | Yes | Yes | Yes |
| Positive News × Content | | Yes | | Yes |
| Positive News × Syntax Controls | | Yes | | Yes |
| Positive News × News Subject FE | | Yes | | Yes |
| Observations | 63,626 | 63,626 | 65,708 | 65,708 |

significantly accelerates information incorporation: while 40%–56% of price adjustment is delayed for low perplexity news, this delay is reduced by 29–52 percentage points for high perplexity news. This is consistent with strain-inducing text triggering deeper, more immediate System 2 processing.

A natural question is whether the perplexity effect varies with firm size. Small capitalization stocks are more prone to underreaction generally (Hou and Moskowitz, 2005), potentially leaving more room for improvement from cognitive triggers. However, small stocks also receive less analyst coverage and institutional attention, so the marginal investor processing news about these firms may differ systematically from those trading large caps. Table 3 presents results separately for large and small capitalization stocks, separated by the median firm size of all articles on each day. The perplexity effect is present in both subsamples with similar magnitudes, though estimated more precisely for small stocks—consistent with the effect operating broadly across the firm size distribution.

### 3.3 News Perplexity and EDGAR Traffic: Underlying Mechanism

To understand the behavioral mechanism through which perplexity mitigates underreaction, I investigate whether strain-inducing text increases deliberate information acquisition. I proxy for costly, active information search using EDGAR traffic, widely validated as reflecting investor information demand (Drake, Roulstone, and Thornock, 2015; Loughran and McDonald, 2017).

The underlying EDGAR server log data is available from 2003–2017 and from 2020 onwards. I restrict analysis to the pre-2017 period, as later data lacks key user agent variables essential for reliable bot identification. To process log files and count human views, I employ the method of Ryans (2017), which eliminates multiple-download counting and provides robust bot classification.

The central hypothesis, derived from dual-process theory, is that increased perplexity

**Table 4** EDGAR Search Traffic Increases with Perplexity

This table presents results from OLS regressions examining the relationship between news perplexity and investor information acquisition, proxied by abnormal EDGAR search traffic on the day of the news release (t=0). The sample consists of 21,004 news events for S&P 500 firms from 2010 to 2017. The dependent variable is Abnormal EDGAR Traffic, calculated as the percentage deviation from the firm's average traffic on the same weekday over the prior 10 weeks, multiplied by 100%. The main independent variables are indicator dummies for moderate perplexity and high perplexity news articles, with low perplexity serving as the omitted baseline category (the constant term reflects the low perplexity group). Abnormal Google Search Volume is calculated similarly to Abnormal EDGAR Traffic, using the Google Ticker Stock Search Volume Index relative to the firm's average traffic on the same weekday over the prior 10 weeks. Control include stock characteristics (volatility, illiquidity, market cap, volume of news coverage), industry fixed effects, content (predicted two-month returns using text embeddings using the methodology of Chen, Kelly, and Xiu, 2022), syntax (article length, Fog-index), news subject fixed effects (earnings, personnel, M&A), and firm fixed effects. Standard errors (in parentheses) are two-way clustered by firm and day. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels.

| | **Abnormal EDGAR Traffic** | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Constant | 13.21*** | 13.53*** | 14.42*** | 18.00*** | |
| | (1.68) | (1.71) | (1.46) | (1.52) | |
| Moderate Perplexity | 8.82*** | 8.60*** | 8.06*** | 4.73*** | 4.19*** |
| | (1.44) | (1.45) | (1.34) | (1.32) | (1.31) |
| High Perplexity | 18.44*** | 17.51*** | 14.67*** | 8.35*** | 6.34*** |
| | (1.89) | (1.88) | (1.76) | (1.80) | (1.71) |
| Abnormal Google Search | | 1.72*** | 2.55*** | 1.96*** | 2.11*** |
| | | (0.33) | (0.27) | (0.25) | (0.24) |
| Stock Controls | | | Yes | Yes | Yes |
| Industry FE | | | Yes | Yes | Yes |
| Content | | | | Yes | Yes |
| Syntax Controls | | | | Yes | Yes |
| News Subject FE | | | | Yes | Yes |
| Firm FE | | | | | Yes |
| Observations | 21,004 | 21,004 | 21,004 | 21,004 | 21,004 |
| Adjusted $R^2$ | 0.00680 | 0.01147 | 0.06512 | 0.12950 | 0.16743 |

induces cognitive strain that disrupts heuristic processing and triggers effortful analysis. This should manifest as increased costly information acquisition. While simple attention or salience might increase initial readership, the System 2 hypothesis specifically predicts increased active, deliberate search for corroborating information in regulatory filings.

I analyze how abnormal EDGAR search traffic on the news release day ($t = 0$) relates

to article perplexity. As a crucial control for general attention (distinct from deliberate analytical search), I use daily Google Ticker Search Volume Index (SVI) from deHaan, Lawrence, and Litjens (2023), matched for S&P 500 stocks and available from 2010.

For both EDGAR and Google traffic, I calculate abnormal search volume as percentage deviation from the firm's average traffic for the same weekday over the preceding 10 weeks, following deHaan, Lawrence, and Litjens (2023). This removes firm-specific seasonality and day-of-week effects.

The final sample comprises 21,004 news events for S&P 500 firms over 2010–2017. I estimate:

$$\text{AbnormalEdgarSearch}_{it} = \beta_{Low} + \beta_{Moderate} I_{Moderate,it} + \beta_{High} I_{High,it} + \gamma' \mathbf{X}_{it} + \varepsilon_{it}, \quad (9)$$

where $I_{Moderate,it}$ and $I_{High,it}$ indicate moderate and high perplexity groups, with low perplexity as the baseline. Controls (centered at sample means) include stock characteristics, industry fixed effects, content, syntax, and news subject fixed effects. Day-of-week fixed effects are excluded as traffic is already benchmarked against same-weekday historical averages.

Table 4 presents results. Column (1) shows raw effects: high perplexity news is associated with a coefficient of 18.44% on abnormal EDGAR traffic. Compared to the baseline constant of 13.21%, this constitutes more than double the abnormal search. Column (2) introduces Google SVI as control. While Google SVI is highly significant, indicating broader attention drives some EDGAR search, the high perplexity coefficient remains large (17.51%) and highly significant. This effect persists after controlling for stock characteristics, content, syntax, and news subjects (Columns 3–4), with the high perplexity coefficient of 8.35% representing 46% of the baseline level (18.00%). Even with firm fixed effects (Column 5), the high perplexity coefficient remains positive and significant at 6.34%, which is 35% of the baseline low-perplexity level in Column (4),

which is the closest specification without firm fixed effects.

**Table 5** EDGAR Search Traffic Increases with Perplexity: Positive vs Negative News

This table extends the analysis of Table 4 by examining whether the effect of perplexity on EDGAR search traffic differs between positive and negative news. The specifications include an indicator for positive news and its interactions with perplexity indicators.

| | **Abnormal EDGAR Traffic** | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Constant | 18.74*** | 18.53*** | |
| | (1.49) | (1.67) | |
| Moderate Perplexity | 4.42*** | 4.37*** | 4.02** |
| | (1.27) | (1.66) | (1.71) |
| High Perplexity | 8.21*** | 9.79*** | 7.56*** |
| | (1.77) | (2.01) | (1.95) |
| Positive News | -0.23 | 0.20 | 0.23 |
| | (0.82) | (2.25) | (2.29) |
| Positive News × Moderate Perplexity | | 0.15 | -0.67 |
| | | (2.46) | (2.51) |
| Positive News × High Perplexity | | -3.15 | -3.17 |
| | | (2.81) | (2.76) |
| Abnormal Google Search | 1.76*** | 1.76*** | 1.85*** |
| | (0.24) | (0.24) | (0.23) |
| Stock Controls | Yes | Yes | Yes |
| Industry FE | Yes | Yes | Yes |
| Content | Yes | Yes | Yes |
| Syntax Controls | Yes | Yes | Yes |
| News Subject FE | Yes | Yes | Yes |
| Firm FE | | | Yes |
| Observations | 21,004 | 21,004 | 21,004 |
| Adjusted $R^2$ | 0.12927 | 0.12932 | 0.16727 |

One might expect the cognitive trigger to operate asymmetrically: investors may be more motivated to verify surprising bad news than good news, or conversely, may engage more deeply with positive surprises that suggest upside opportunity. Table 5 examines whether the perplexity effect on EDGAR search differs between positive and negative news. The interaction terms are statistically insignificant, indicating that strain-inducing text prompts deliberate information acquisition regardless of news direction.

**Table 6** Perplexity Effect on EDGAR Search Traffic: Interaction with General Attention

This table extends the analysis of Table 4 by examining whether the effect of perplexity on EDGAR search traffic varies with general attention. The specifications add interactions between perplexity indicators and Abnormal Google Search Volume Index.

| | **Abnormal EDGAR Traffic** | |
|---|---|---|
| | (1) | (2) |
| Constant | 18.60*** | |
| | (1.50) | |
| Moderate Perplexity | 4.44*** | 3.73*** |
| | (1.27) | (1.27) |
| High Perplexity | 7.88*** | 5.68*** |
| | (1.77) | (1.67) |
| Abnormal Google Search | 1.62** | 1.64** |
| | (0.67) | (0.63) |
| Abnormal Google Search × Moderate Perplexity | -0.08 | 0.02 |
| | (0.72) | (0.70) |
| Abnormal Google Search × High Perplexity | 1.29 | 1.30 |
| | (0.83) | (0.80) |
| Stock Controls | Yes | Yes |
| Industry FE | Yes | Yes |
| Content | Yes | Yes |
| Syntax Controls | Yes | Yes |
| News Subject FE | Yes | Yes |
| Firm FE | | Yes |
| Observations | 21,004 | 21,004 |
| Adjusted $R^2$ | 0.12950 | 0.16743 |

A potential concern is that perplexity simply proxies for salience—perhaps unusual text captures attention broadly, and EDGAR search merely follows general interest. If so, the perplexity effect should be amplified when general attention is already elevated. Table 6 tests this by interacting perplexity indicators with abnormal Google Search Volume. The interactions are insignificant, suggesting that perplexity's effect on EDGAR search operates independently of broad attention levels. This supports the interpretation that strain-inducing text triggers deliberate analytical effort rather than merely riding waves of public interest.

**Table 7** EDGAR Search Traffic and Traditional Readability (Fog Index)

This table replicates the analysis of Table 4, replacing perplexity with the Fog Index as the measure of traditional readability.

| | **Abnormal EDGAR Traffic** | |
|---|---|---|
| | (1) | (2) |
| Constant | 25.76*** | |
| | (1.23) | |
| Moderate Fog | -2.96** | -1.41 |
| | (1.19) | (1.13) |
| High Fog | -1.76 | -0.18 |
| | (1.57) | (1.51) |
| Abnormal Google Search | 1.78*** | 1.87*** |
| | (0.24) | (0.23) |
| Stock Controls | Yes | Yes |
| Industry FE | Yes | Yes |
| Content | Yes | Yes |
| Syntax Controls | Yes | Yes |
| News Subject FE | Yes | Yes |
| Firm FE | | Yes |
| Observations | 21,004 | 21,004 |
| Adjusted $R^2$ | 0.12848 | 0.16671 |

Finally, if perplexity captures cognitive strain, one might ask whether traditional readability measures produce similar effects. Table 7 replicates the analysis using the Fog Index in place of perplexity. The coefficients on moderate and high Fog are statistically insignificant and, if anything, negative—the opposite of what a generic "cognitive strain" story would predict. This null result further distinguishes narrative unpredictability (captured by perplexity) from traditional readability (captured by Fog index), reinforcing that the two dimensions of cognitive strain operate through distinct channels.

These results strongly support the hypothesized cognitive mechanism. Strain-inducing text significantly increases deliberate EDGAR search even after controlling for general attention. Investors do not merely notice high-perplexity text; the cognitive effort required prompts them to engage in effortful information acquisition, presumably to verify or better

understand the narrative. This active search behavior provides a micro-foundation for the main asset pricing result: deeper processing triggered by cognitive strain leads to more complete, efficient information incorporation, thereby mitigating underreaction.

# 4 Conclusion

This paper investigates whether strain-inducing text in financial news—independent of its core informational content—can improve investor information processing and enhance market efficiency. The prevailing view in finance holds that cognitive strain, typically proxied by low readability, hinders processing and harms market outcomes. Drawing on dual-process theory, I propose an alternative: strain-inducing text can function as a beneficial cognitive trigger, disrupting investors' heuristic-based "System 1" intuition and engaging their deliberate, analytical "System 2" faculties.

To test this hypothesis, I measure strain-inducing text using perplexity, a metric from time-aligned large language models (ChronoGPT) that quantifies narrative unpredictability. Analyzing a sample of high-impact, firm-specific news events from the Dow Jones Newswires, I find strong support for my hypothesis. The central finding is that strain-inducing text significantly mitigates market underreaction. High perplexity news exhibits substantially faster and more complete price reaction, with the delayed response ratio falling from 40% to 11%—a 29 percentage point reduction. This result is robust to controls including traditional readability (Fog Index) and news subject fixed effects, demonstrating that perplexity captures a distinct economic channel.

I provide direct evidence for the proposed cognitive mechanism by examining investor behavior. High news perplexity is positively associated with abnormal EDGAR search traffic on the news release day—an incremental increase about half the baseline low-perplexity level—even after controlling for general attention via Google search volume.

This suggests strain-inducing text prompts costly, effortful information acquisition—an indication of System 2 engagement—rather than merely capturing fleeting attention.

This study makes three contributions. First, it challenges the uniformly negative view of cognitive strain in finance by identifying a beneficial form: strain induced by narrative unpredictability. It provides the first evidence that text-induced cognitive strain can enhance market efficiency, specifically showing that unpredictable narratives mitigate underreaction. Second, it conceptualizes and measures narrative unpredictability via LLM perplexity as a dimension of text distinct from both informational content and syntactic complexity. Third, it demonstrates a novel, theory-driven application of LLMs in finance, using an intrinsic model property to measure a cognitive construct and explain a canonical market anomaly.

The findings also open avenues for research on causal mechanisms. Future work could combine this asset-pricing approach with laboratory or field experiments that exogenously manipulate strain-inducing properties of text while holding content fixed, directly measuring belief updating, trading behavior, and information acquisition under different cognitive constraints. Such work would sharpen the micro-foundations of the System 1-to-System 2 channel and clarify when cognitive strain improves market outcomes. Ultimately, this research demonstrates that *how* a story is told—specifically, its capacity to induce cognitive strain—is a critical, and previously unmeasured, factor in market efficiency.

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